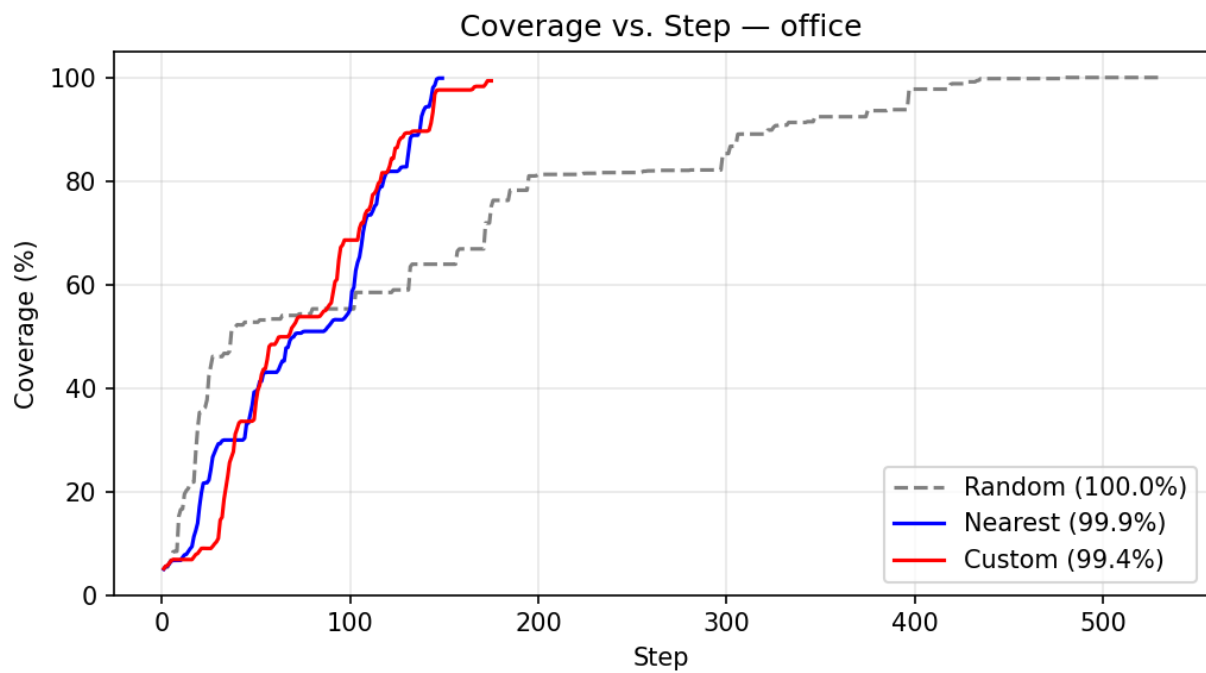
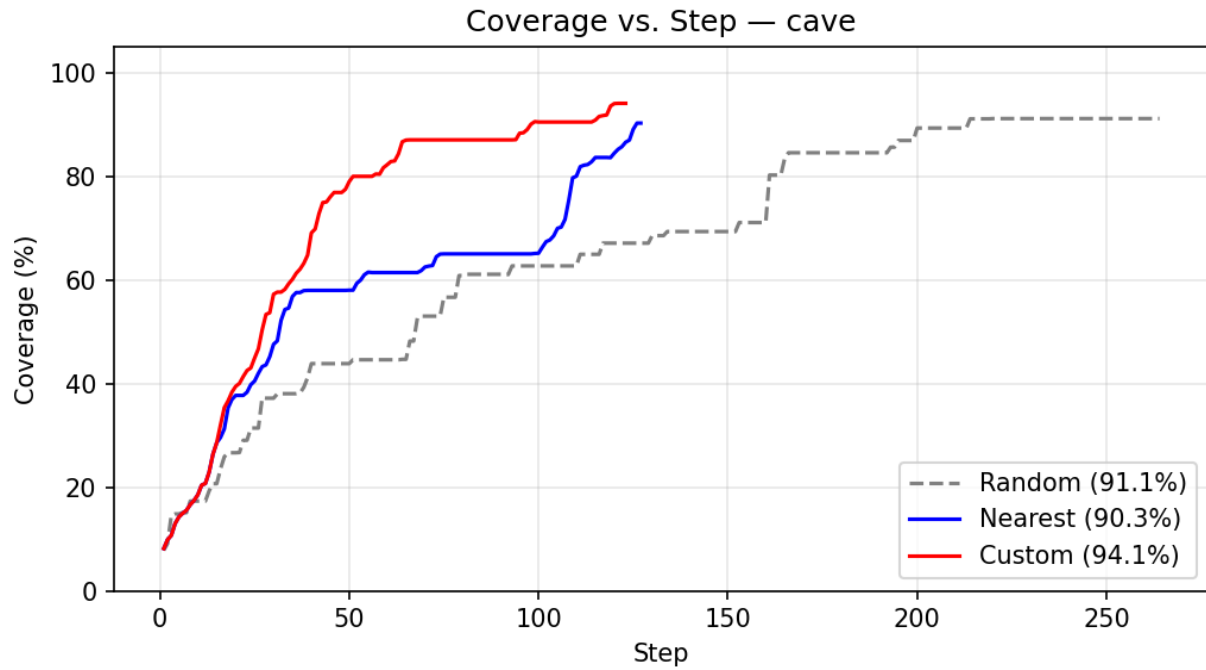
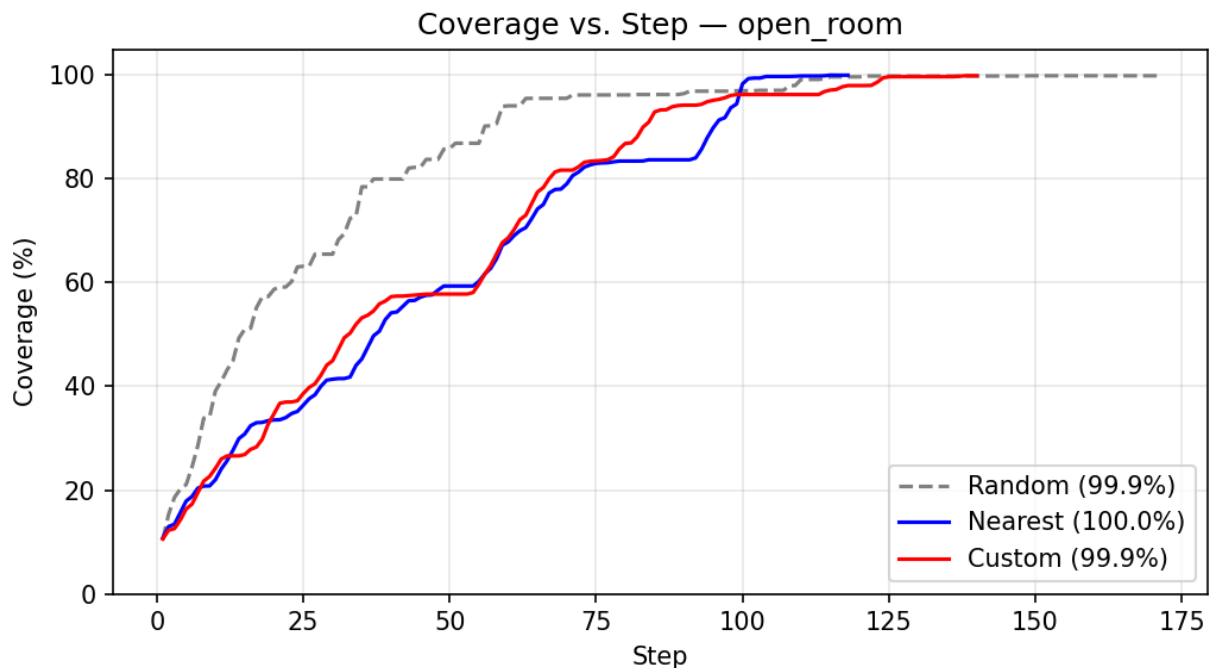






All three strategies achieve high coverage eventually in open-room and office maps, but they differ in exploration speed. The random baseline requires more steps than the other two in cave and office because randomly selecting free cells cause it to revisit already-explored areas.

By contrast, the nearest-frontier converges faster in office because the room-to-room structure lets the robot naturally sweep through connected rooms in order. However, it seems to struggle on the cave map narrow passages create small frontiers near the robot while the larger areas lie further away. This setup causes this strategy to chase small frontier fragments that are closer to the robot while the larger areas remain unexplored.

My custom strategy accounts for the size of the unexplored region with `nearby_unknown_count`. This balances cost with expected information gain. This strategy outperforms others on the cave map because it tends to explore larger regions despite potentially higher cost. However, on other maps where all frontiers are roughly equally informative, this strategy provides no benefit.